

TrES-3b

R-0354

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250605 · created 2026-07-18T11:09:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460831.7853 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.019
Transit depth (Rp/Rs)^2	3.85 +/- 0.76 [%]
Orbital Inclination (inc)	81.73 +/- 0.65
Airmass coefficient 1 (a1)	1.035 +/- 0.015
Airmass coefficient 2 (a2)	-0.028 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#3 - [409, 433]
Optimal Aperture	4.49
Optimal Annulus	9.26
Transit Duration (day)	0.057 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.019 vs 0.16309 (deviation 1.68σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.05σ)
Mid-time O-C	-3.2 min
Depth SNR	10.0
Residual scatter	1.44 %

Light curve

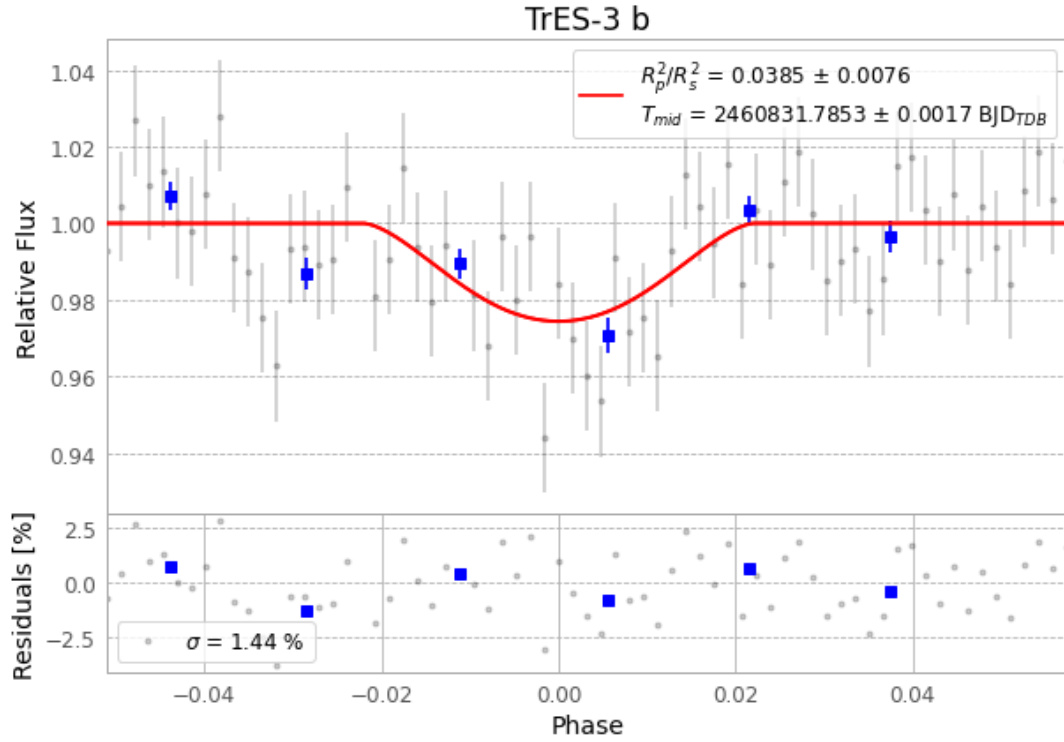


Figure 1 — FinalLightCurve_TrES-3 b_2025-06-04.png (SHA-256 9f76c387f3e94850...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 218], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7a717fe8c132e78e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

69 FITS frames (45 MB), TRES-3250605050916.FITS → TRES-3250605083315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250605.log (030564c4946c...), FinalLightCurve_TrES-3 b_2025-06-04.png (9f76c387f3e9...), FinalLightCurve_TrES-3 b_2025-06-04.csv (6133bf3eb041...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 11:09Z.